

Team 2026

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NYC Weather & Traffic

Objective

Build a predictive model that uses real-time weather data (rain, temperature) to forecast NYC traffic delays. Deliver live Tableau dashboards that city planners can use to anticipate congestion and adjust signal timing or transit schedules.

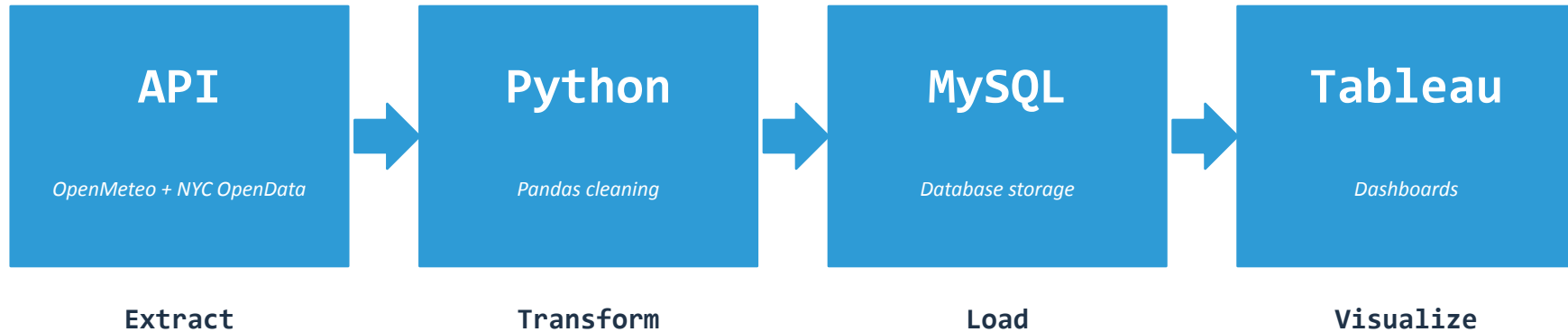
- How much do rain intensity and temperature shifts change estimated delay time across NYC boroughs — and which conditions create the sharpest jumps?
- Is there a measurable precipitation tipping point where delays spike, and does it differ by borough?
- Does adding weather features to a traffic volume baseline meaningfully improve delay prediction — or does prior traffic history already explain most of the variance?

What Problem Our Model Solves

- Goal: predict how rain and temperature affect **estimated delay time** across NYC boroughs
- Target variable: traffic volume (or volume drop vs baseline)
- Predictors: precipitation, temperature, humidity, day-of-week, hour-of-day
- Regression captures both linear trends and rush-hour amplification

Data Pipeline

Extract → Transform → Load → Visualize



Data Sources

01

Open Meteo API

Historical & real-time weather

Key fields

Precipitation, temperature, humidity, wind speed

02

NYC OpenData

Traffic speed & volume

Key fields

Borough, segment, hour, traffic volume by date

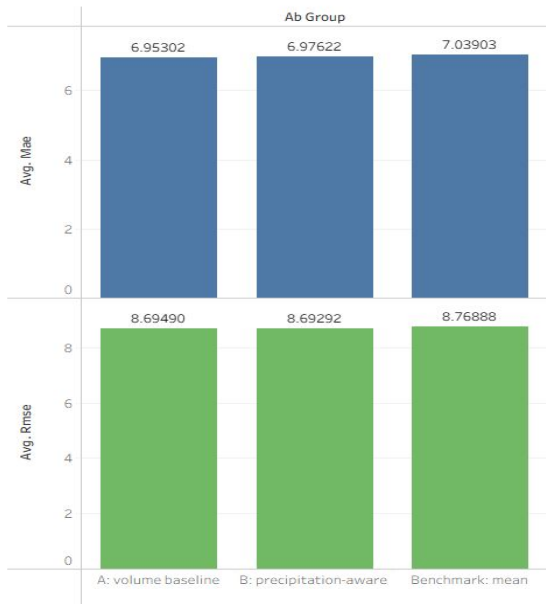
Regression Model

Regression Model Logic

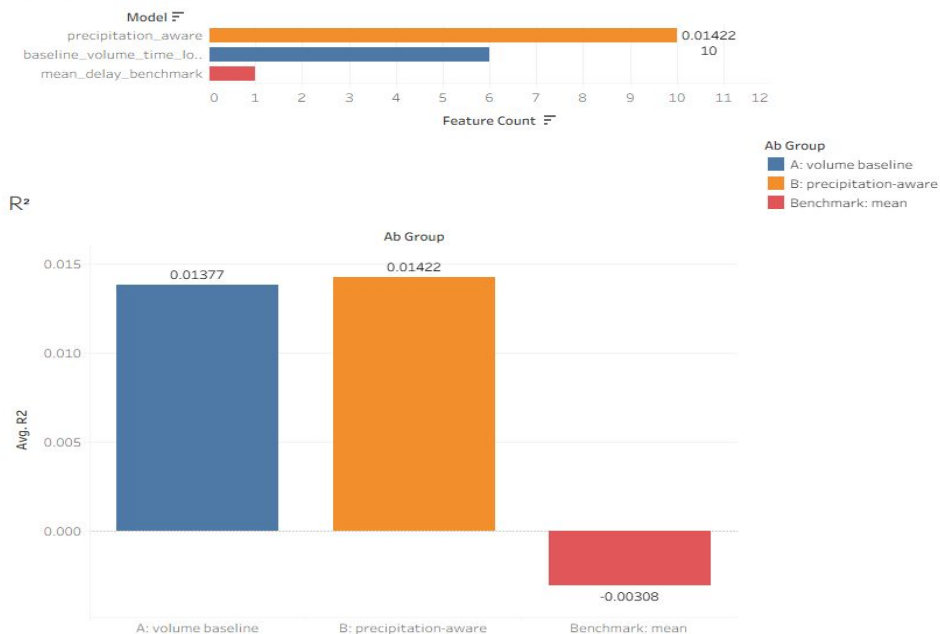
- Models the relationship between weather features and estimated delay minutes
- Multivariate version adds day-of-week and hour-of-day to capture commuting patterns.
- Trained on historical data, evaluated using R^2 and RMSE on held-out test sets.
- Produces per-condition delay predictions rather than a single overall average.

A/B Test: Model Performance

MAE, RMSE



Model Accuracy vs Features



Side-by-side accuracy: weather-only baseline vs weather + traffic history

Why This Matters

- Heavy rain constantly drops traffic volume below baseline
- Temperature swings compound delays in winter and during heat waves
- Weather effects multiply during peak rush hour windows
- Boroughs respond differently to the same conditions

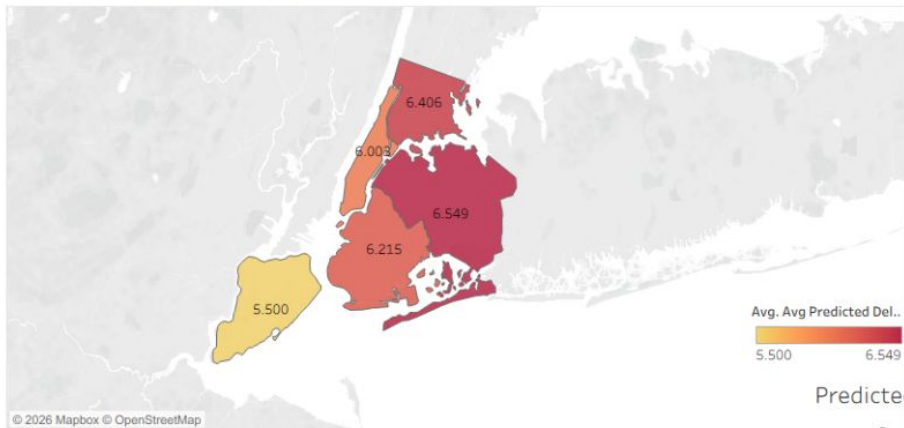


Visualizations

Tableau dashboard outputs



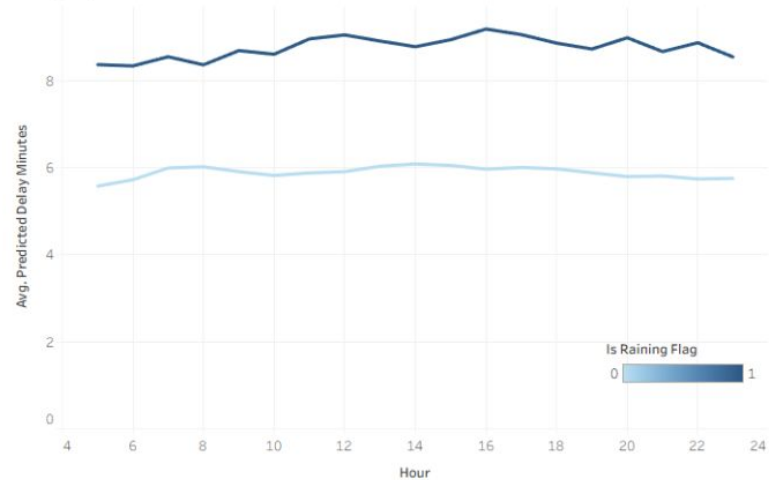
Predicted Delay By Borough



Predicted Delay



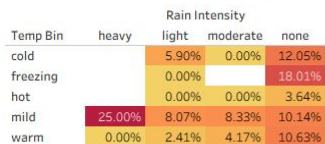
Delay by Hour



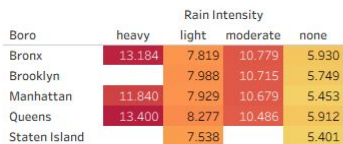
Predicted Delay VS Actual Delay



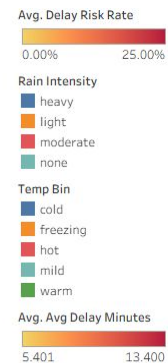
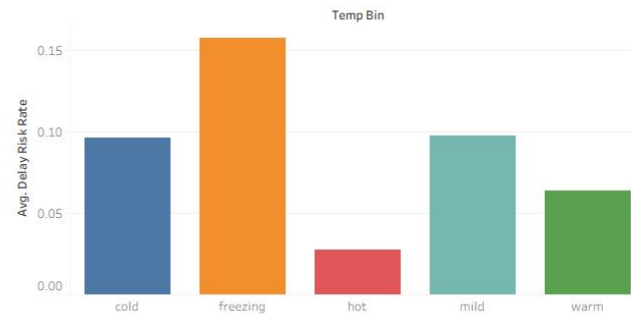
Combined Weather Heatmap



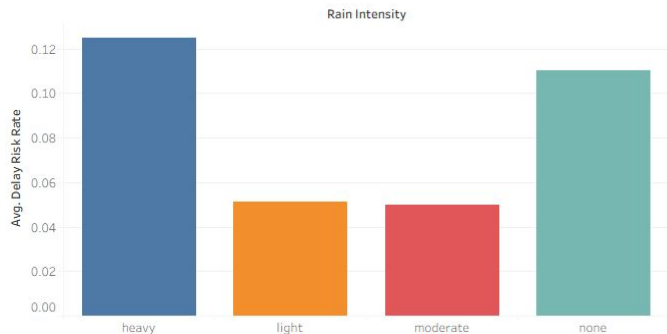
Rain Intensity by Borough Heatmap



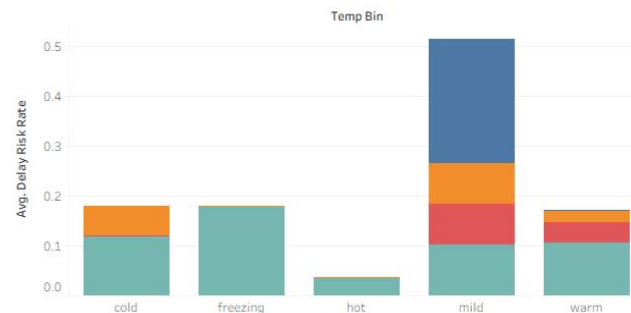
Temperature vs Delay Risk



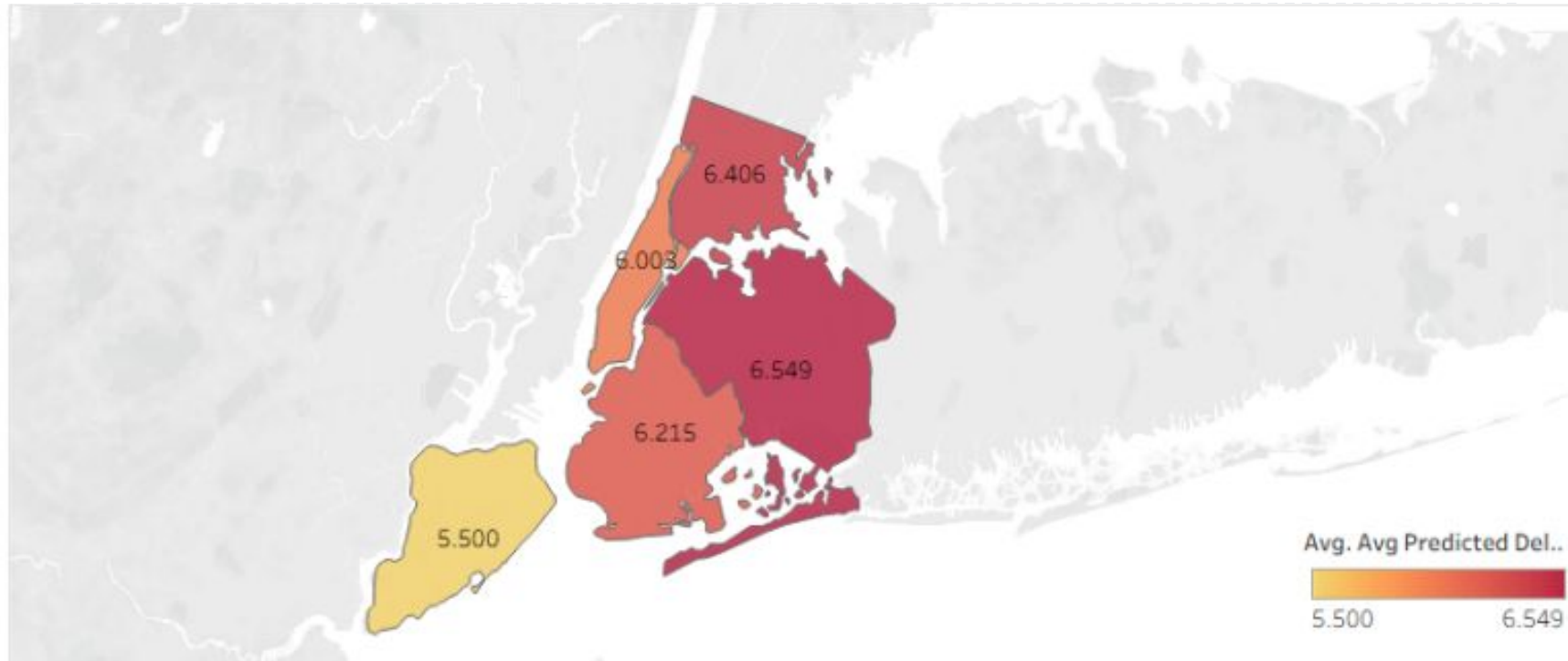
Rain vs Delay Risk



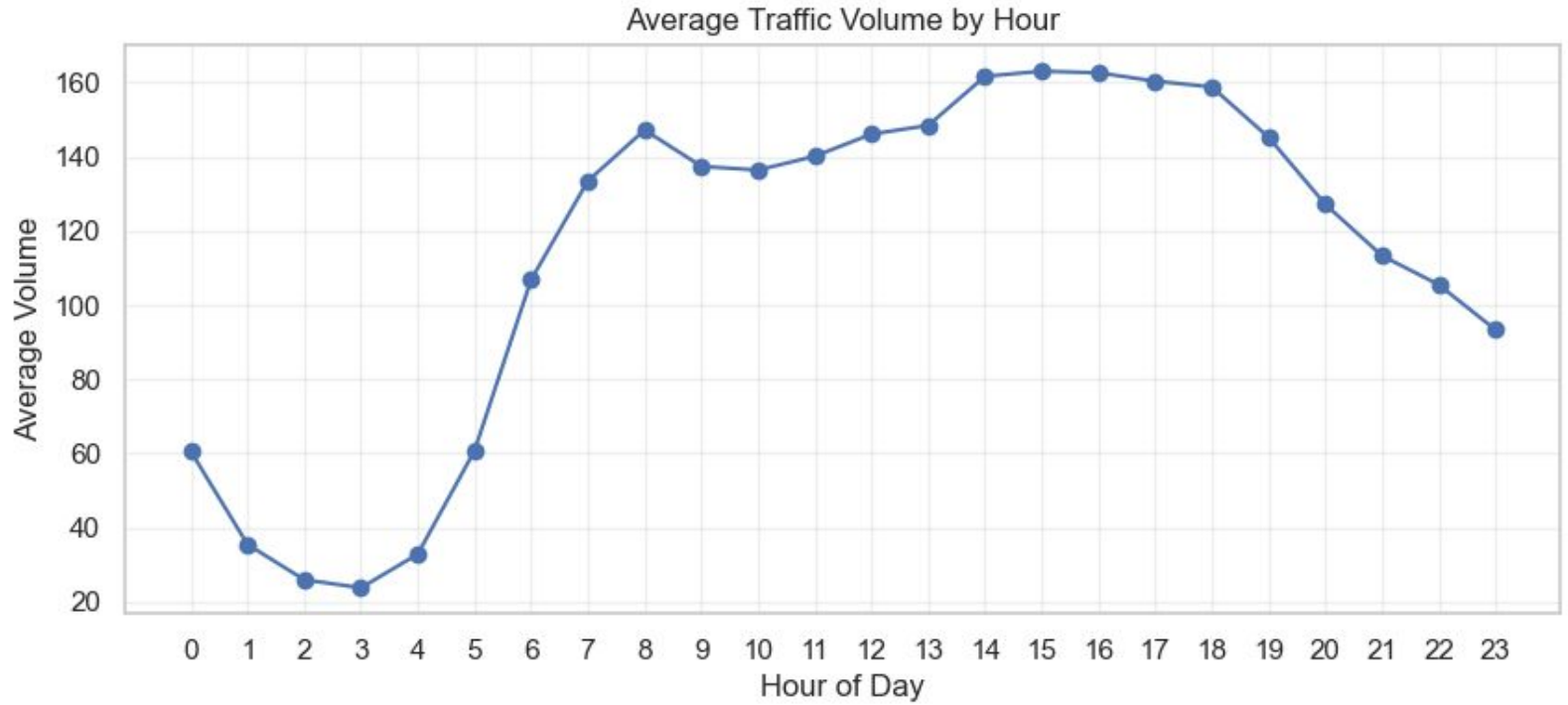
Weather Condition Delay Risk



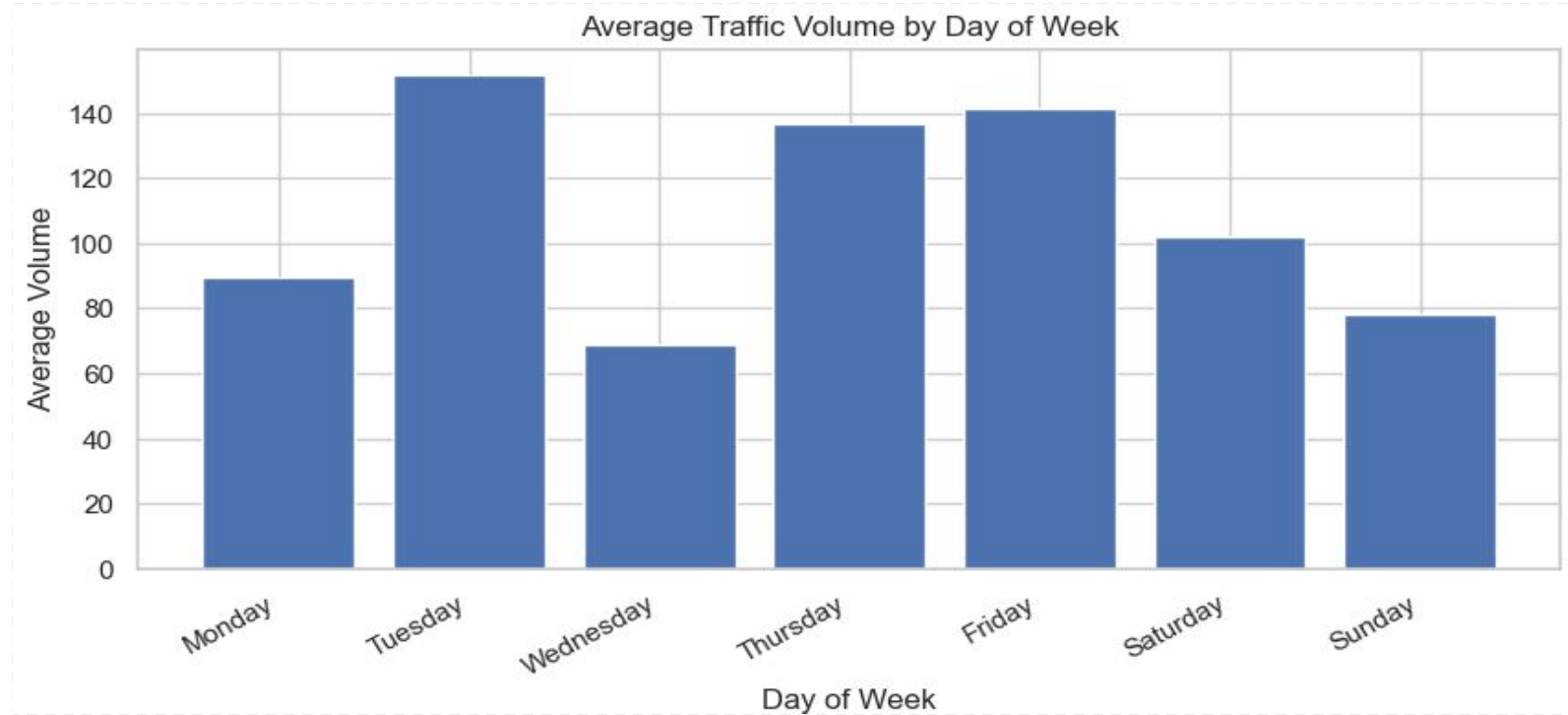
Predicted Delay by Borough



Average Traffic by Hour

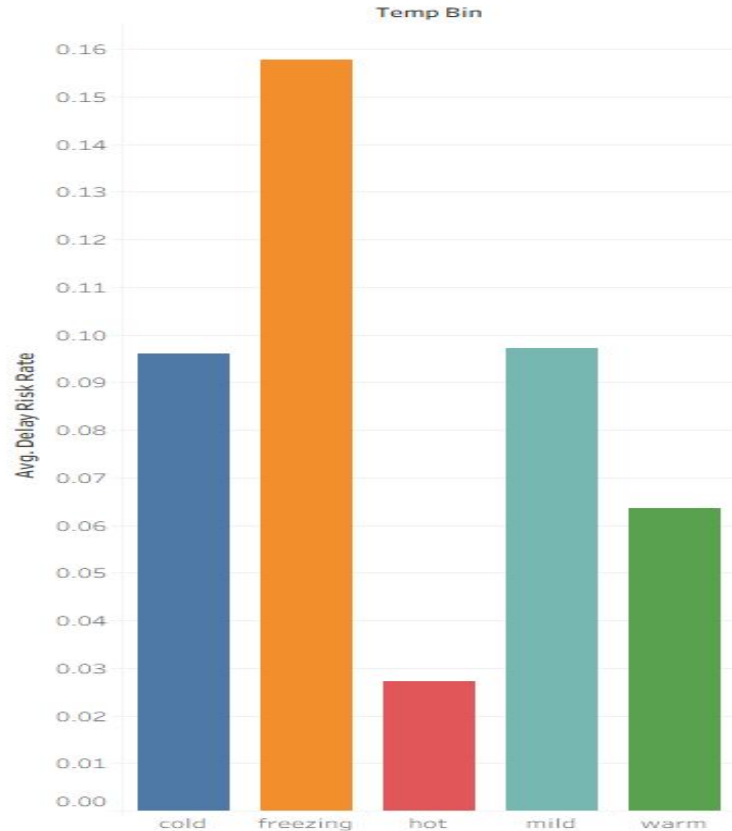


Day-of-Week Breakdown

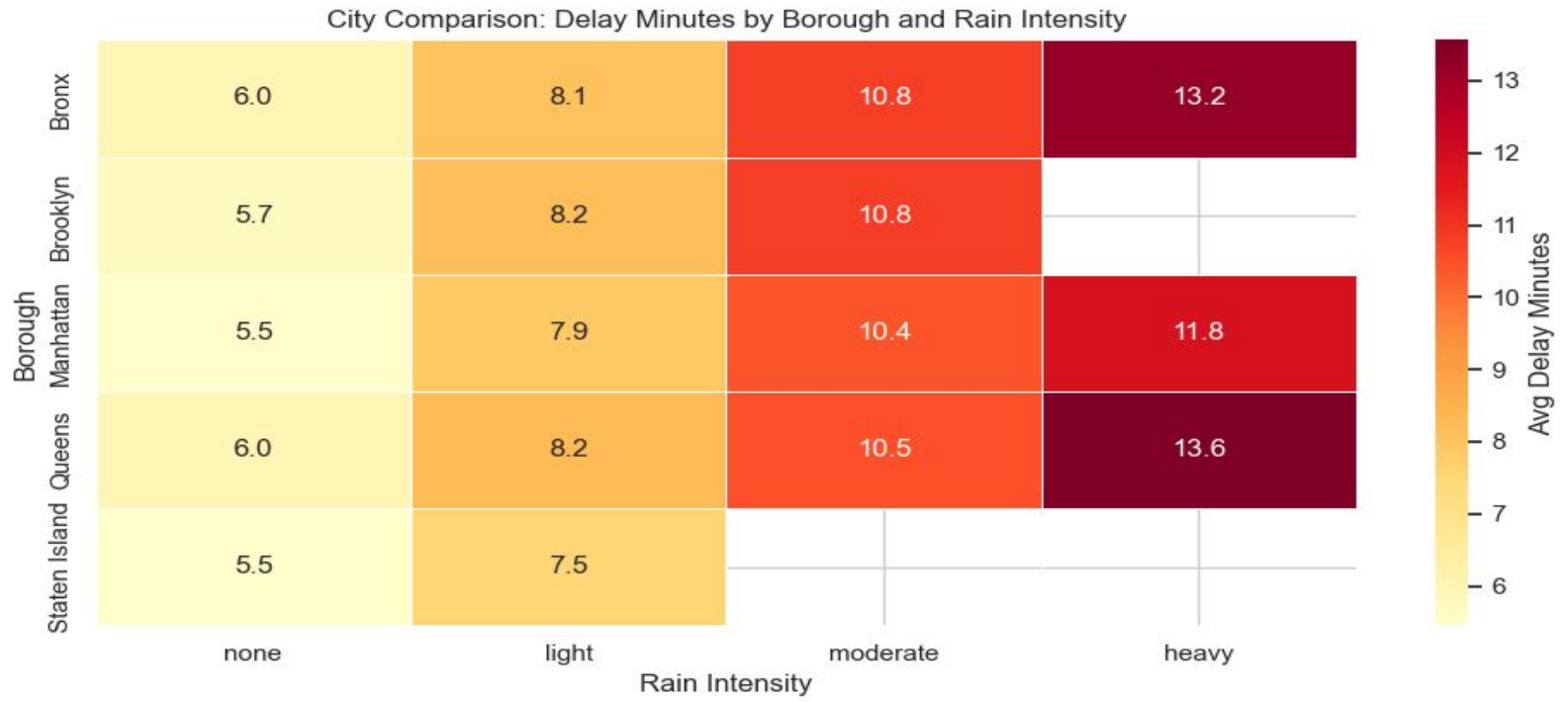


Temperature Impact on Volume

Temperature vs Delay Risk



Borough and Rain Density / Delay Minutes



Rain Intensity and Temp / Borough vs Delay Risk Rate

Rain Intensity by Borough
Heatmap

Boro	Rain Intensity			
	heavy	light	moderate	none
Bronx	13.184	7.819	10.779	5.930
Brooklyn		7.988	10.715	5.749
Manhattan	11.840	7.929	10.679	5.453
Queens	13.400	8.277	10.486	5.912
Staten Island		7.538		5.401

Combined Weather Heatmap

Temp Bin	Rain Intensity			
	heavy	light	moderate	none
cold		5.90%	0.00%	12.05%
freezing		0.00%		18.01%
hot		0.00%	0.00%	3.64%
mild	25.00%	8.07%	8.33%	10.14%
warm	0.00%	2.41%	4.17%	10.63%



Business + Recommendation

Our prediction strategy and visualization prep



Business Impact & Recommendations

Who Benefits

- Commuters — 13+ min delays in Queens/Bronx during rainy rush hour; plan ahead or take transit
- City Planners — Borough heatmaps pinpoint where infrastructure investment matters most
- NYC DOT — Live dashboard flags high-risk windows before congestion peaks

Recommendations

- Alert DOT when precipitation exceeds 0.06 inches during rush hours in Queens and Bronx
- Deploy precipitation-aware model for live Tableau predictions citywide
- Flag Queens and Bronx specifically during evening rush (4–9 PM) where weather + volume effects compound the most

Key Findings

- Heavy rain doubles delays: ~5.8 min (dry) → ~13.2 min (heavy rain)
- Rush hour compounds this further — peak hour heavy rain pushes 14–16 min
- Delay risk accelerates past 0.06 inches; moderate and heavy rain have similar impact
- Queens and Bronx hit hardest; Staten Island least affected
- Volume and location drive most variance — weather adds meaningful signal on top

Conclusion

Next Steps

- We answered the question: rain and temperature measurably impact NYC delays
- Our precipitation-aware model outperforms the baseline — weather data adds real value
- Expand weather features to include snow, ice, and wind for more variables.
- Add real-time 311/incident feed to capture non-weather delay causes
- Build toward a borough-level congestion intelligence system for the NYC DOT

Thank you

Team 2026 — Questions?